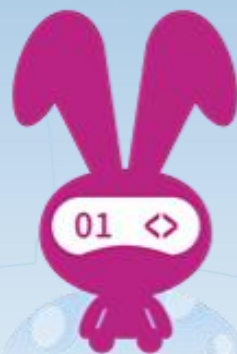




MoonBit月兔



AI时代下的云原生开发平台

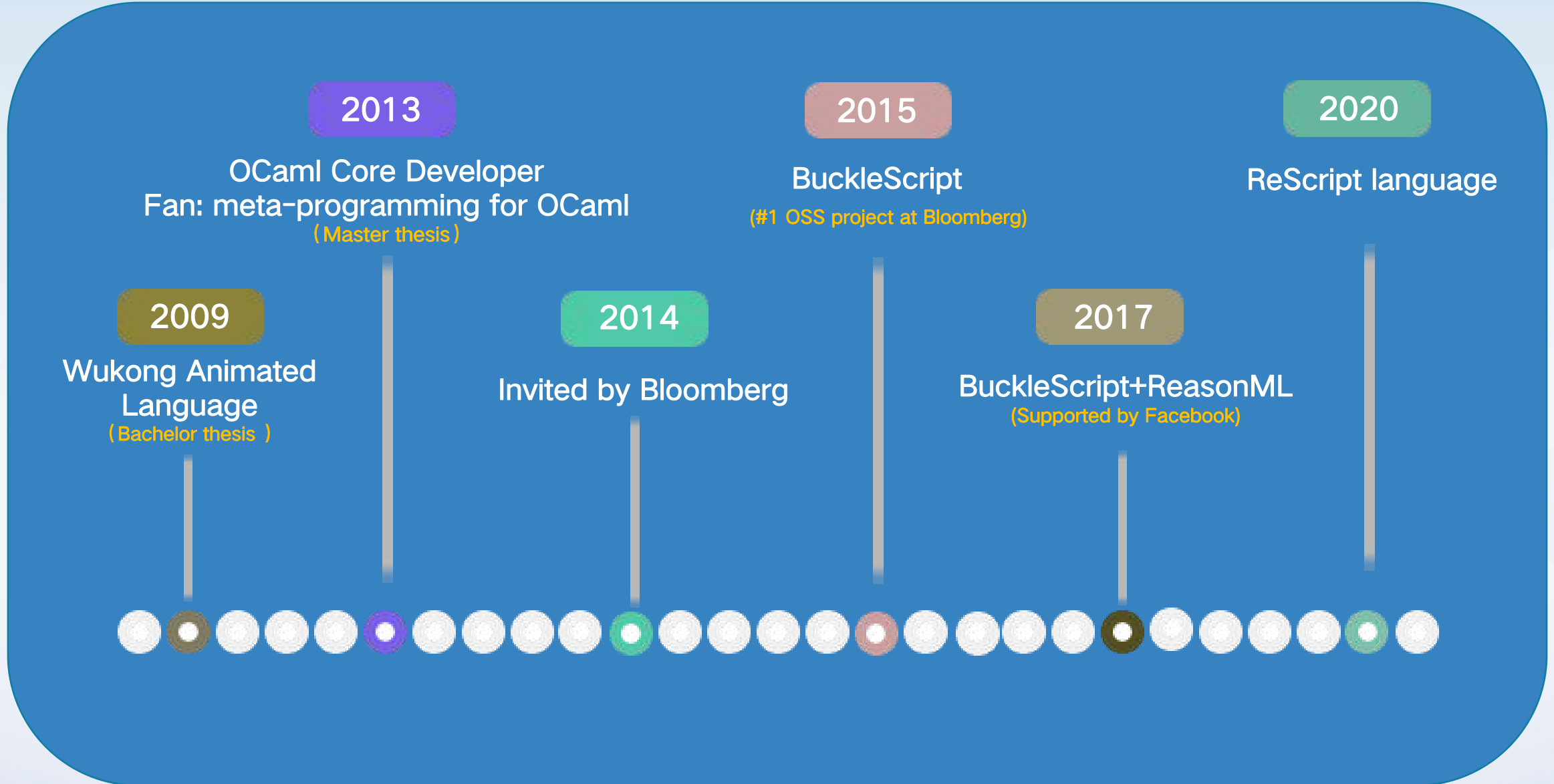


About Me

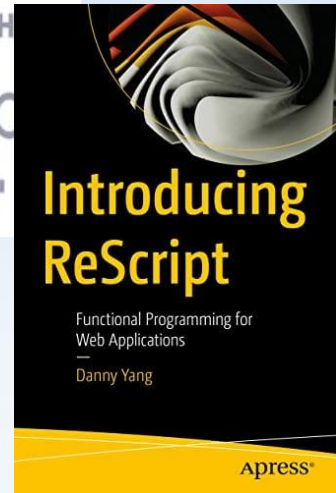
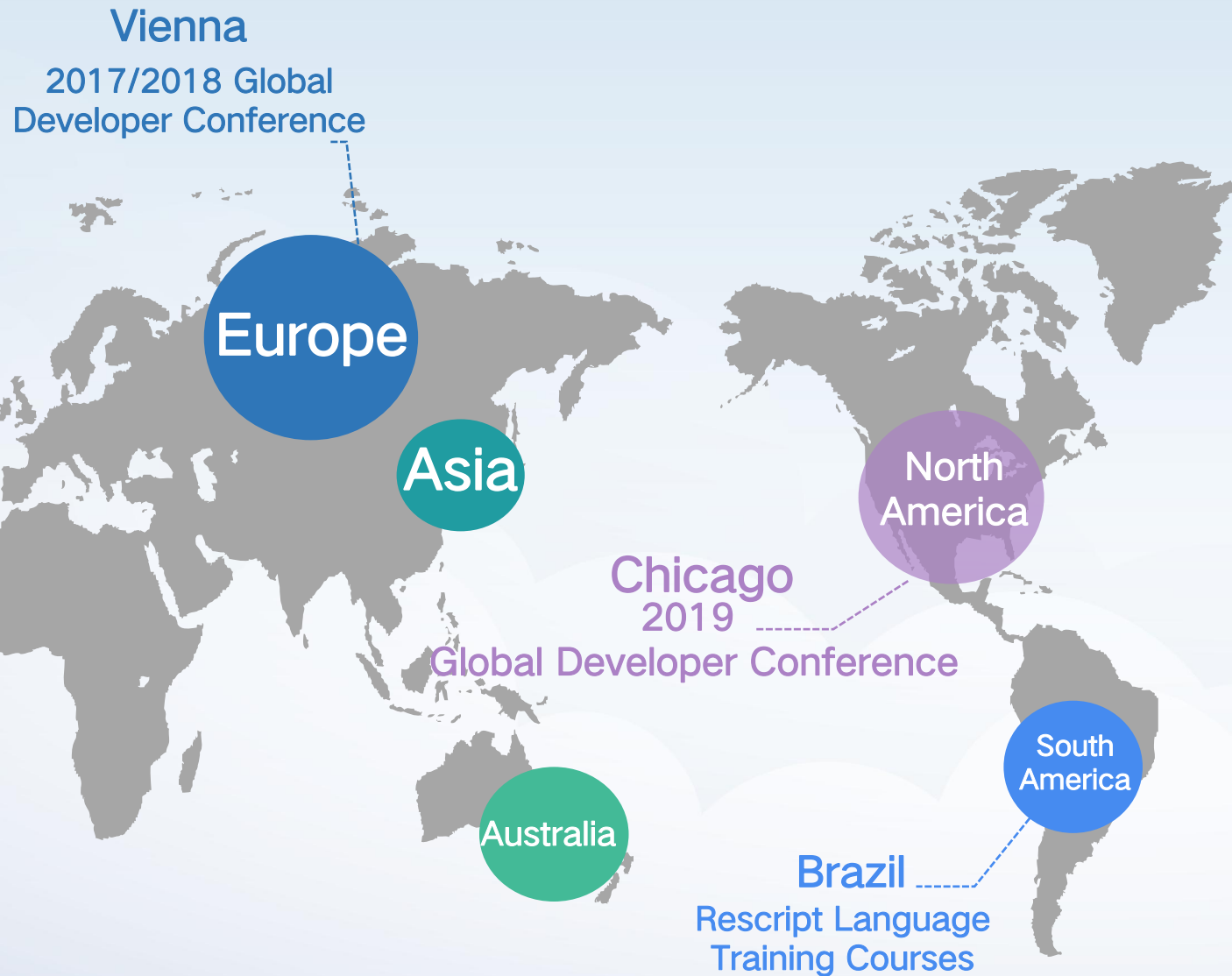
张宏波
Hongbo Zhang

- Lead of MoonBit Platform
- Chair Scientist@IDEA
- OCaml Core Developer
(2013 as a core developer; 2023 OCaml received the SIGPLAN award)
- Author of ReScript
(was BuckleScript&ReasonML)

Timeline



ReScript Community



01 Supported by community contributors, **major enterprise adoption, version upgrades, operations** no longer require the direct involvement of original author

02  **Paul Biggar**
@paulbiggar **Founder of CircleCI/Darklang**

.@rescriptlang is a good language. Arguably the best compiles-to-JS language. Tl

Takeaway from ReScript

What does rescript compiler give up by compiling so fast?

 **zeroexcuses** Feb 13

The rescript compiler appears to compile faster than anything with similar type complexity, i.e. Rust, Jsso, Scala, Haskell. (esbuild feels faster, but esbuild mostly just strips away TS type signatures).

What is the rescript compiler giving up by compiling so fast? What is being sacrificed ?

 **Hongbo**  Team 11d

I think we did not give up too much compared with alternatives.

The performance mostly came from **vertical integration**, when the build system, compiler, library comes from a single mind with performance in mind, it will be significantly faster than alternatives. It is a pity most language developers don't care performance *that much*.

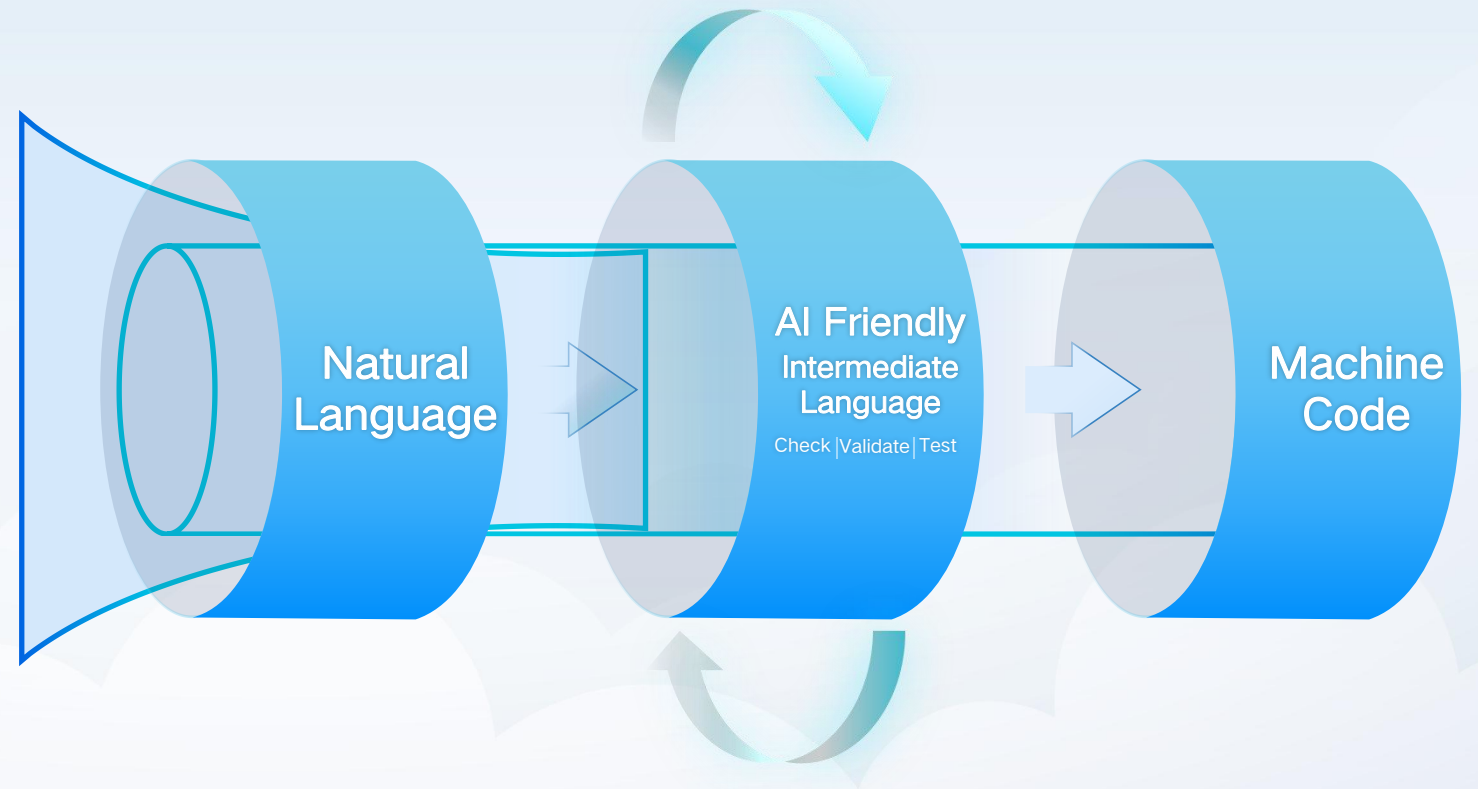
I am toying around a new language with wasm backend support, it only took 28ms (cold start) to type check 6 packages (no parallelism employed yet). Modern CPUs are fast to do lots of things within milliseconds.



Vertical Integration

Future Trends: AI+Cloud

Now: Human+Machine
Future: Human+AI+Machine



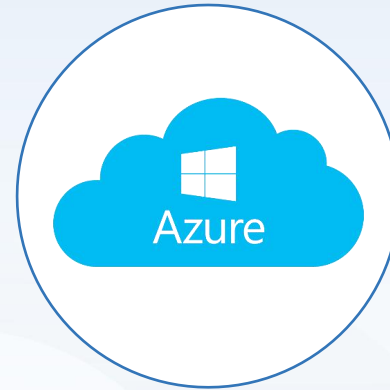
Future Trends: AI+Cloud



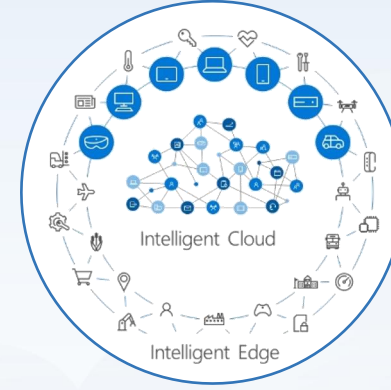
Mainframe



Personal Computer



Intelligent Cloud



Intelligent Cloud+Intelligent Edge



MoonBit

AI时代下的云原生开发平台

MoonBit 08/18 Announcement

 **Hacker News**
new | past | comments | ask | show | jobs | submit

1. ▲ Moonbit: Fast, compact and user friendly language for WebAssembly (moonbitlang.com)
111 points by dlib 5 hours ago | hide | 56 comments
2. ▲ How to Roman Republic 101, Part IIIb: Imperium (acoup.blog)
103 points by Khaine 7 hours ago | hide | 24 comments
3. ▲ New motherboard improves old CRT television (hackaday.com)
113 points by rbanffy 9 hours ago | hide | 38 comments
4. ▲ Kernighan and Pike were right: Do one thing, and do it well (medium.com/source-and-buggy)
88 points by yarapavan 6 hours ago | hide | 70 comments
5. ▲ 2009scape (2009scape.org)
196 points by OsrsNeedsf2P 11 hours ago | hide | 88 comments
6. ▲ Comparing Windows uninstallers, getting upset and then deciding to make my own (jv16powertools.com)
16 points by thunderbong 2 hours ago | hide | 5 comments

 **Hacker News**
new | past | comments | ask | show | jobs | submit login

1. ▲ Moonbit: Fast, compact and user friendly language for WebAssembly (moonbitlang.com)
111 points by dlib 5 hours ago | hide | 56 comments

2023/8/18 MoonBit first Announcement, reaching the Top of *HackerNews

*HackerNews: One of the most influential tech communities in Silicon Valley.

 **Rob Palmer**
@robpalmer2

🌟 @moonbitlang is a new Wasm-first language by @bobzhang1988

- 🌐 Optimized for tiny Wasm output
- ⚡ Performance "comparable to Rust"
- 👉 User-friendly, e.g. exhaustive pattern matching
- 🗑️ Managed memory (WasmGC in-progress)
- 🔧 CLI & IDE tooling
- ⚠️ In Beta

moonbitlang.com

*Rob Palmer: JavaScript Infrastructure & Tooling at Bloomberg. Co-chairing @TC39

 **Hongbo Zhang**
@bobzhang1988

Excited to introduce @moonbitlang for Cloud and Edge computing! 🌟 Build as fast as @golang. 🌟 Run as fast as @rustlang. 🌟 Tiny size of Wasm output. Plus, start with a great cloud IDE right out of the box! moonbitlang.com/blog/first-ann... #WebAssembly #EdgeComputing #CloudComputing

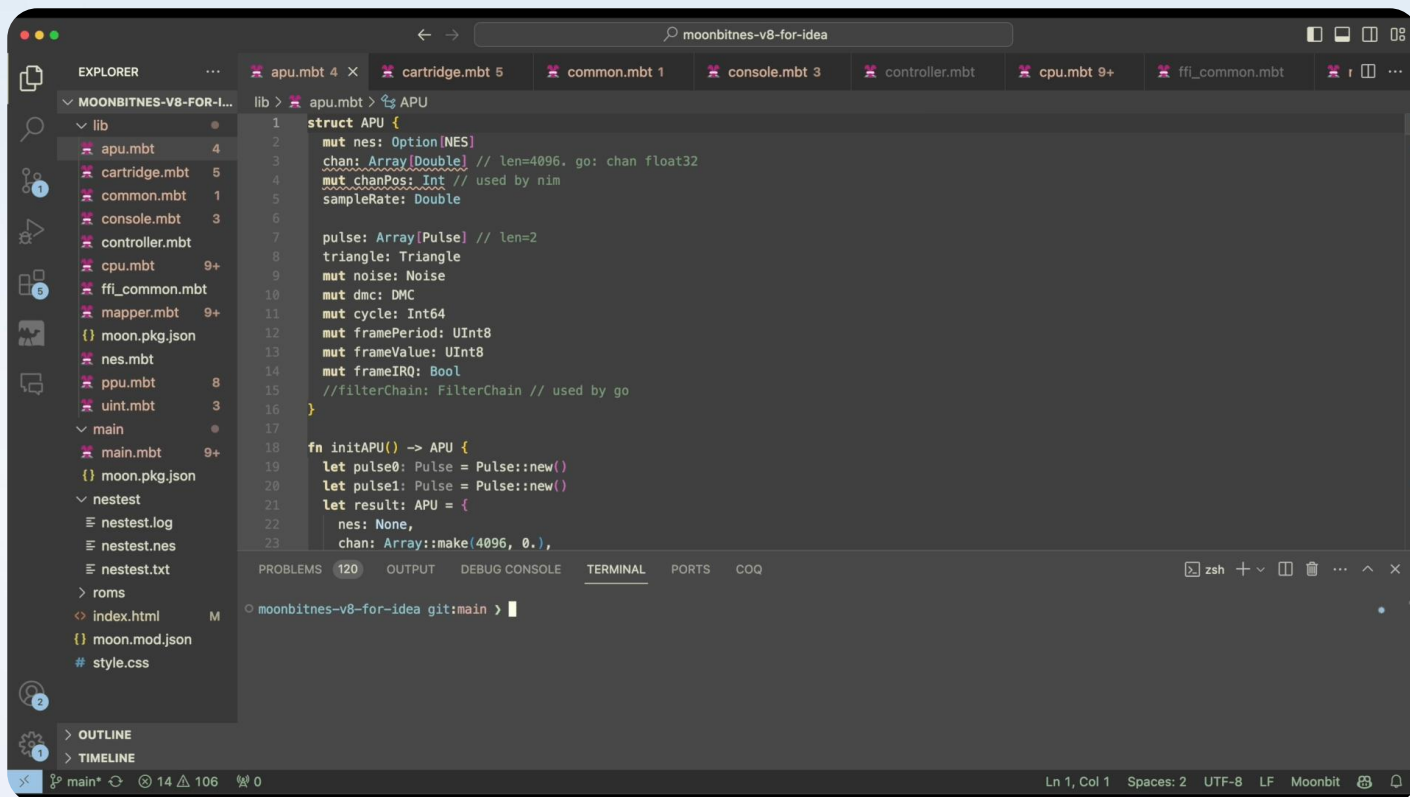
5:01 PM · Aug 18, 2023 · 31.5K Views

 5  47  156  11 

*Twitter post garnered over 30,000 views.

MoonBit早期用户

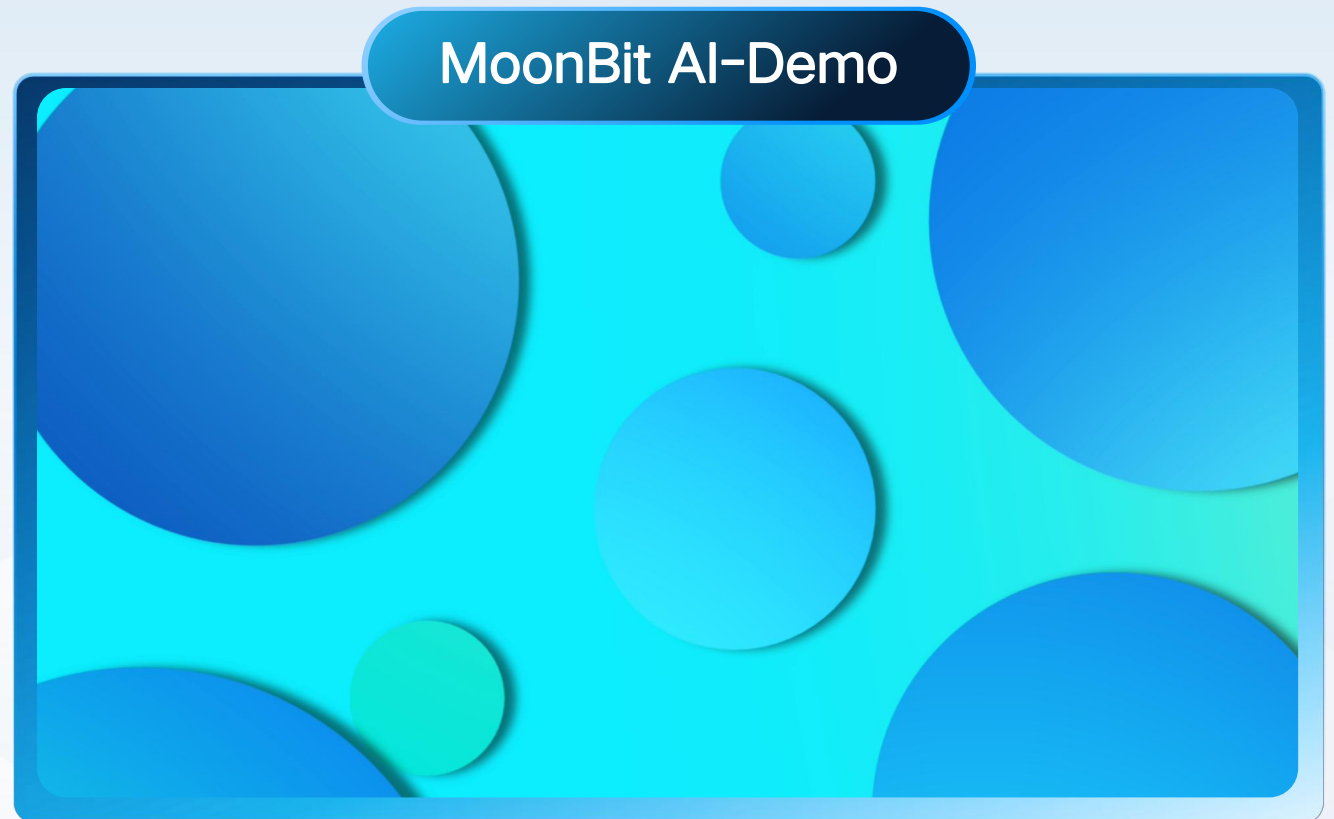
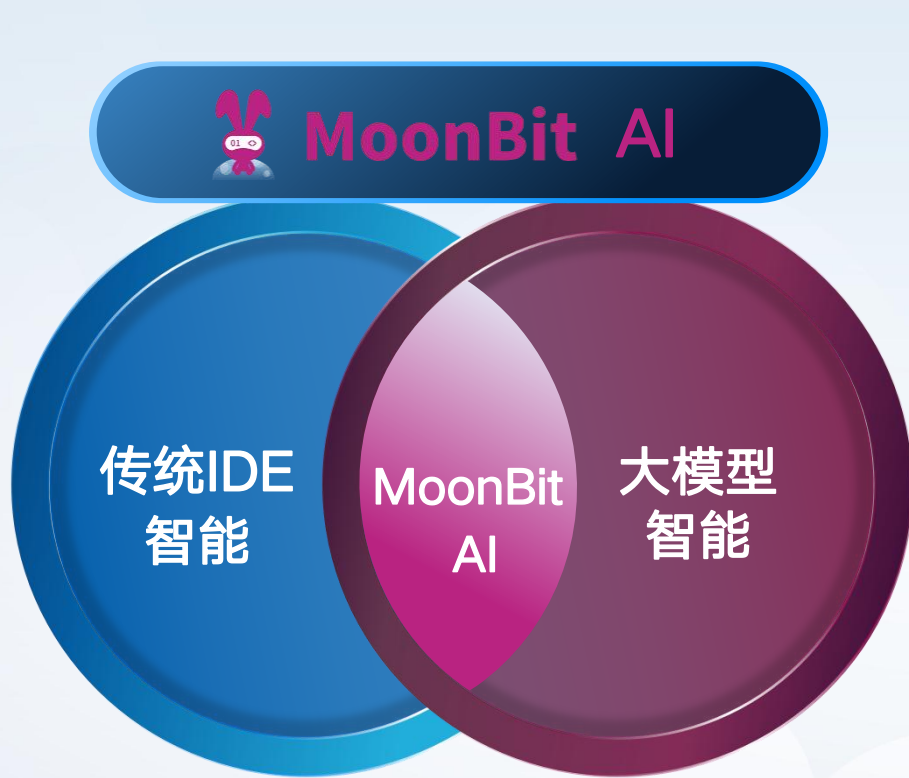
刘汝佳使用MoonBit实现的Nes模拟器



刘汝佳 / 前国际信息学
奥林匹克竞赛国家队教练

“MoonBit是我见过的第一个从一开始就严肃的统筹和平衡诸多元素的语言：语法优美性、实用性、编译速度、代码体积、后端生态和与AI的交互等。再加上宏波本人在PL方面的造诣、经历和极大的热情，我对MoonBit抱有很大的期待。”

MoonBit AI



MoonBit AI

```
/// A simple 2-D geometry library

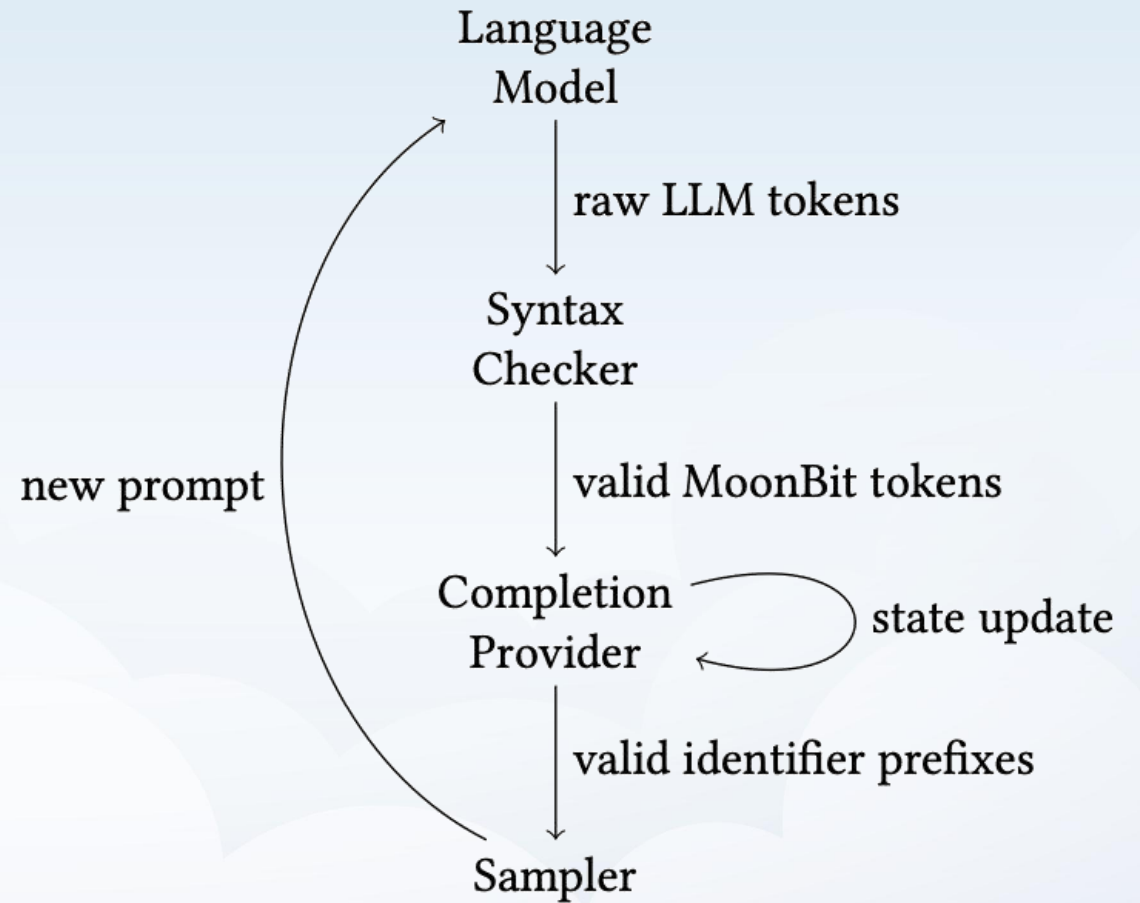
// Computes Euclidean distance between two points.
pub fn distance(p1: Point, p2: Point) -> Double {

  /// All content below is invisible to our large language model.

  /// A 2-D Point.
  pub struct Point {
    coord_x: Double
    coord_y: Double
  }

  /// Use FFI to take the square root of a double.
  pub fn take_sqrt(self: Double) -> Double = "math" "sqrt"
```

MoonBit AI-Semantics Based Sampling



MoonBit AI

▶ 局部重采样

```
pub [
  x
  y
  fn
```

▶ 全局重采样

```
pub fn sum (p:Point) {
```

```
  p. [
    x
    y
    co
```

```
  struct Point {
    cordx:Int
  }
```

Three Waves of Cloud Computing



Virtual Machines



Container



WebAssembly



WebAssembly



WA

Scenario Applications

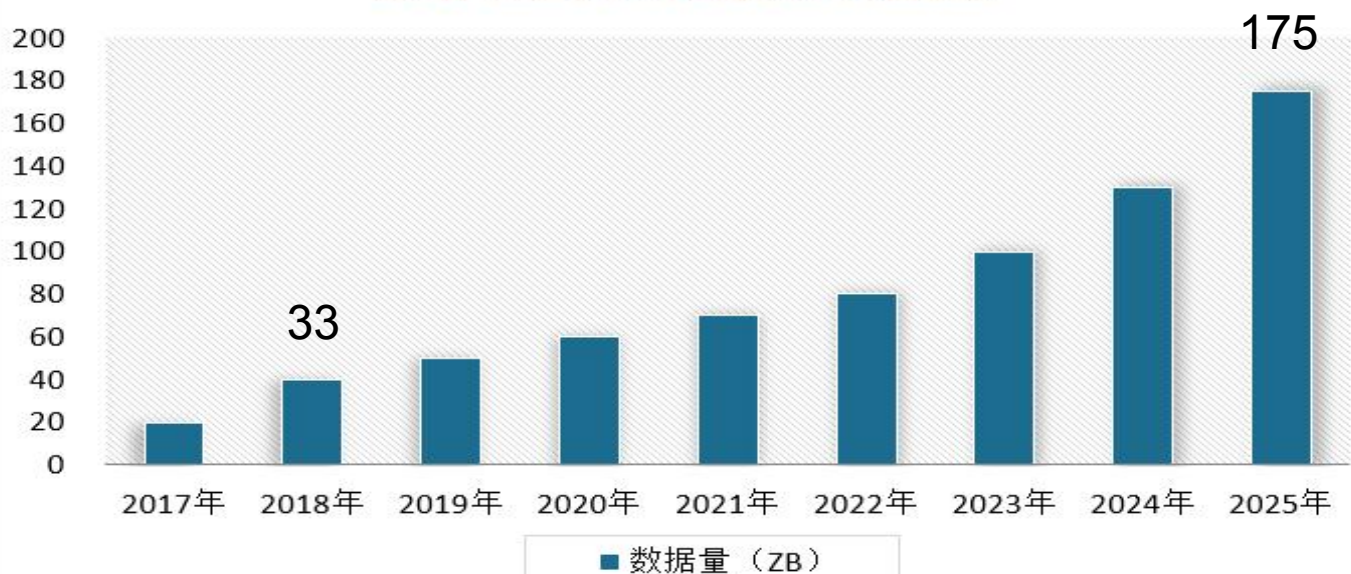


```
(module
  (func $fib (export "fib") (param i64)
    (result i64)
    (if (result i64) (i64.le_u
      (local.get 0) (i64.const 1))
      (then (i64.const 1))
      (else
        (i64.add
          (call $fib (i64.sub
            (local.get 0) (i64.const 2)))
          (call $fib (i64.sub
            (local.get 0) (i64.const 1)))
        )
      )
    )
  )
)
```

WebAssembly

Data
Explosion

2017-2025年全球数据量增长预测



Security

Performance,
Latency & Power
consumption

Software
OTA
Upgrade

Heterogeneous
Platforms

WASM Replacing Traditional Solutions: Lua/JS/Native

- Over **1000 times** faster than Lua
- More portable than JS, as many microcontroller units **have less than 100k of memory**
- Exceptionally stable, reducing crash rates **from 1 in 1000 to zero**

W
A
S
M

High Performance

Portable

Multilingual

Post MVP

WASM



WebAssembly

SUPPORTED LANGUAGES

FIRST-CLASS CITIZENS

C, C++, Rust, Zig, TinyGo,
Solidity, Fe, AssemblyScript,
Grain, Motoko

SECOND-CLASS CITIZENS

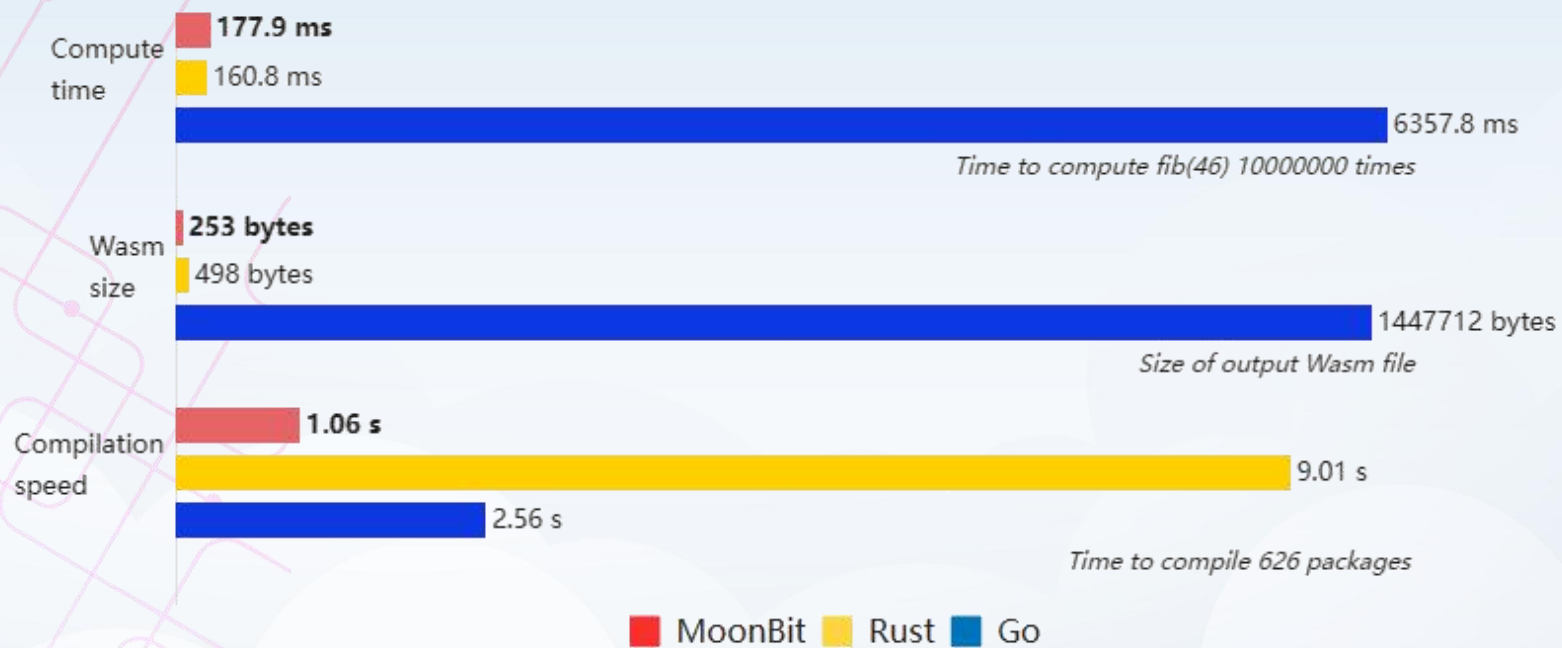
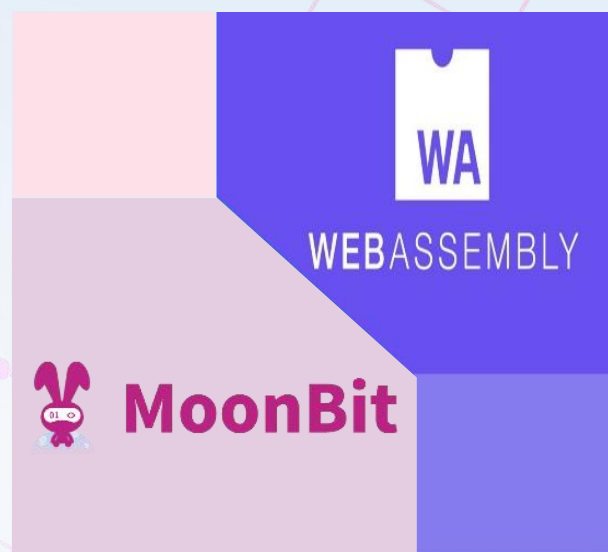
JavaScript, C#, F#, Go,
Python, Java, Kotlin,
Swift, D, Pascal

"An assembly-like binary format for code that can be executed by web browsers. It can be generated from higher level languages like C++ and Rust. The idea is that it runs a lot faster than JavaScript."

ACM SIGPLAN introduces Wasm

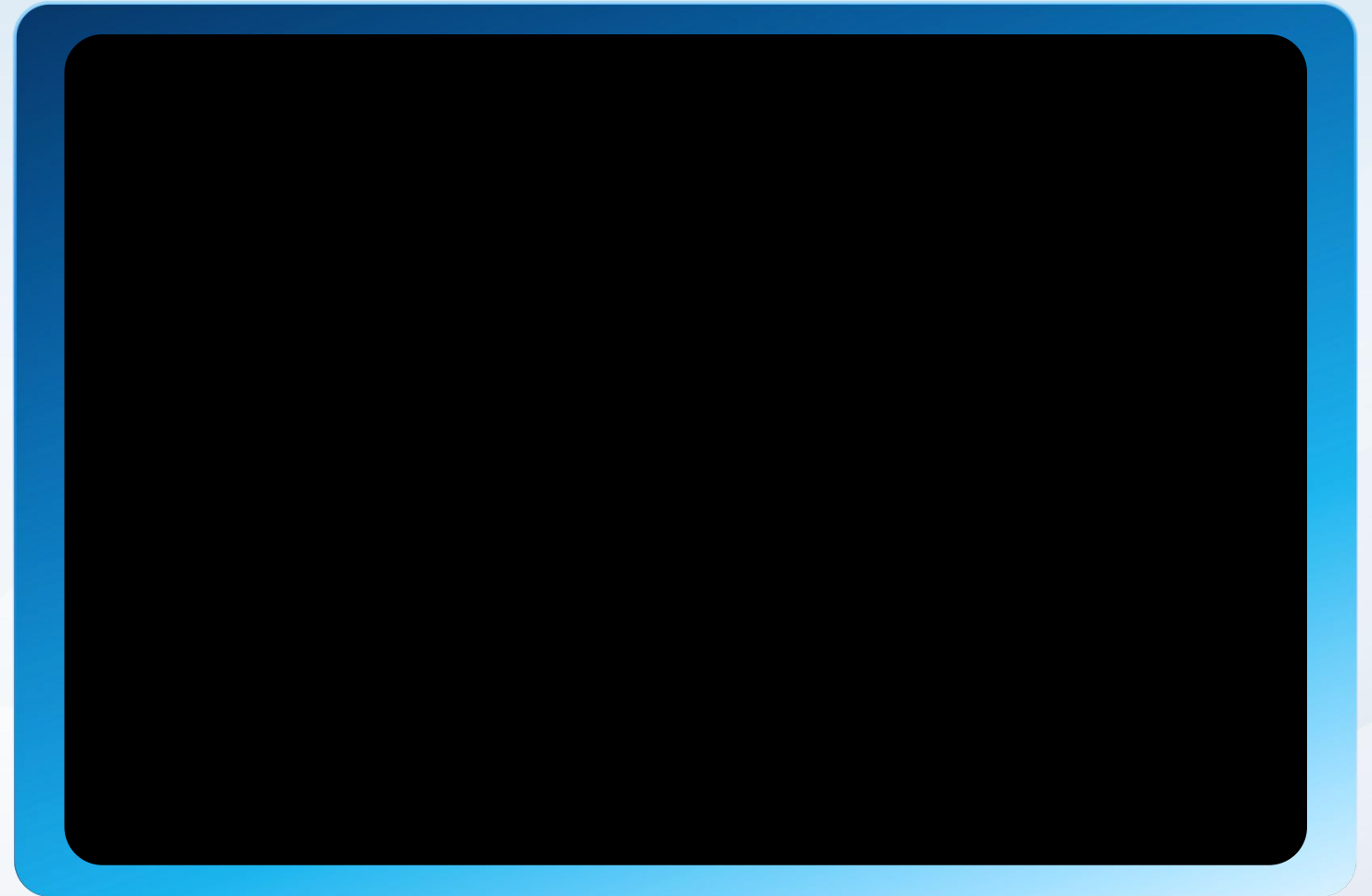
Out of the top 20 languages in the Redmonk ranking, 17 can run on Wasm

MoonBit WASM



MoonBit Platform

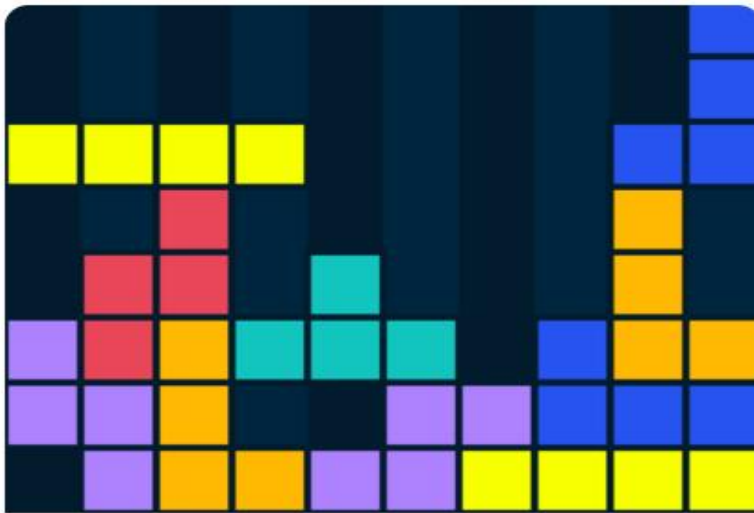
- ▶ MoonBit IDE（集成开发环境）
 - ▶ MoonBit Debugger（调试器）
 - ▶ MoonBit Live Coding（实时编码）
-



MoonBit Gallery

MoonBit Gallery

Welcome to the MoonBit gallery.



Tetris

A traditional tetris game development in MoonBit using web canvas API. You can change the MoonBit code and see the result on the fly.



Mario

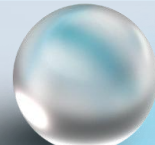
A mario game development in MoonBit using web canvas API. You can change the MoonBit code and see the result on the fly.

Try: moonbitlang.cn/gallery

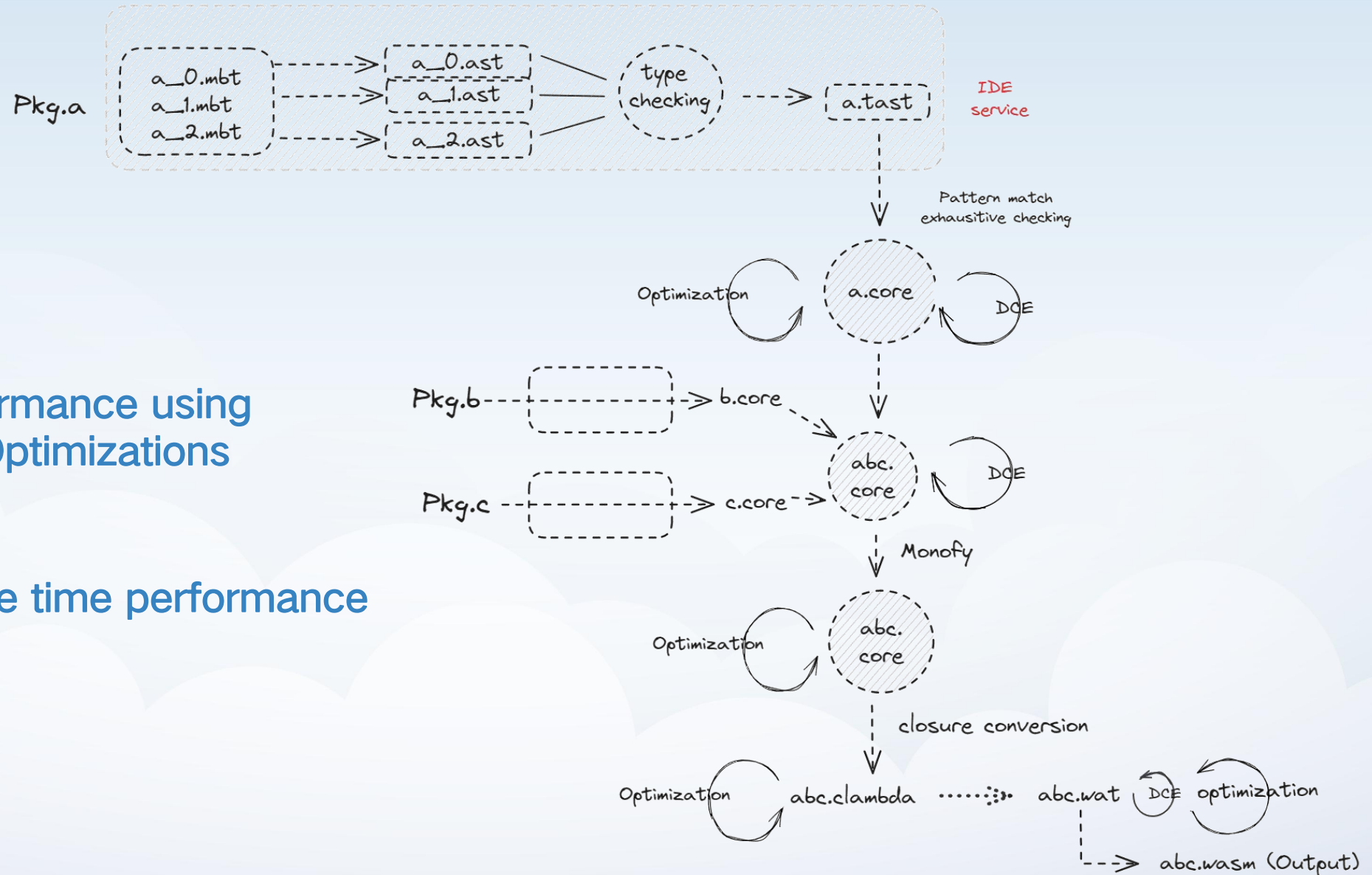
MoonBit CLI



```
> moon new hello
> cd hello
> cat main/main.mbt
fn init {
    @lib.hello().print()
}
> moon run main
Hello, world!
```



MoonBit Compiler Architecture



- Fast runtime performance using Whole Program Optimizations
- State of art compile time performance

Key design decisions in MoonBit



MoonBit in Education



No need to install



Zero-install modern IDE:

Real-time responsiveness

Optimal user experience: Not dependent on network latency.



Programming philosophy

Modern programming principles

Higher-order functions

pattern matching

functional programming

type systems

modular programming

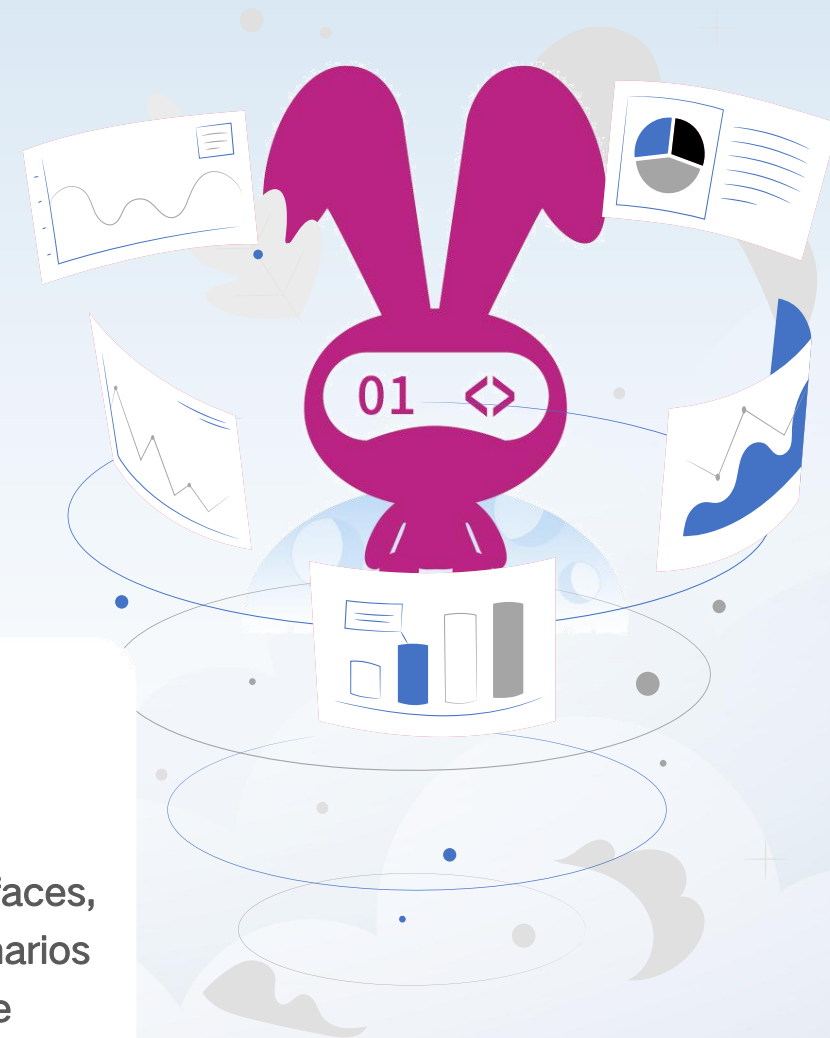
Better
Scheme@MIT



Fun & practical

Fun: Rich graphical interfaces, versatile application scenarios

Practical: Industrial-grade ecosystem



MoonBit Online Public Lecture

《现代编程思想=数据结构+算法+模块化编程》

我的视频

40

最新发布

最多播放

最多收藏

▶ 播放全部

更多 >



新手程序员公开课 | 现代编程思想4: 多元组、结构体与枚举类型

▶ 680

11-23



新手程序员公开课 | 现代编程思想3.2: 递归与动态规划

▶ 711

11-17



新手程序员公开课 | 现代编程思想3.1: 函数列表与递归

▶ 506

11-14



新手程序员公开课 | 现代编程思想2.3: 月兔中的表达式

▶ 679

11-9



新手程序员公开课 | 现代编程思想2.2: 月兔中的表达式

▶ 777

11-7



新手程序员公开课 | 现代编程思想2.1: 月兔开发环境的准备

▶ 1114

11-3



新手程序员公开课 | 现代编程思想1.0: 课程介绍与程序设计

▶ 3339

10-31



程序员节特别策划 | 预告片上线! 《现代编程思想》公开课

▶ 1475

10-24

MoonBit CLI

```
/// and returns a new avl tree node with appropriate restructuring if necessary.
36 func bal[U](l: T[U], v: U, r: T[U]) -> T[U] {
37   let hl: Int = l.height()
38   let hr: Int = r.height()
39
40   // Left subtree is taller by more than 2 level
41   if hl > hr + 2 {
42     match l {
43       Empty =>
44         Empty // impossible
45       Node(l1, lv, lr, _) =>
46         if l1.height() >= lr.height() {
47           create(l1, lv, create(lr, v, r))
48         } else {
49           match lr {
50             Empty =>
51               Empty // impossible
52             Node(lr1, lrv, lrr, _) =>
53               create(create(l1, lv, lr1), lrv, create(lrr, v, r))
54           }
55         }
56     }
57   } else if hr > hl + 2 {
```

✓ Pattern matching

Path:

[Try.moonbitlang.com](https://try.moonbitlang.com) 012_avl_tree.mbt

MoonBit Roadmap

2024.2



Beta Preview

2024.6



Beta
Ready for industrial
use cases

2025



Feature
freeze

Q&A

Forum:

<https://taolun.moonbitlang.cn/>

<https://discuss.moonbitlang.com/>

Official Site:

<https://www.moonbitlang.com/>

<https://www.moonbitlang.cn/>



Find Us

MoonBit Official WeChat



Thanks.

